

An Incidence-Kernel Bound for Minimal Counterexamples to the Erdős–Gyárfás Conjecture

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Abstract

The Erdős–Gyárfás conjecture asserts that every finite graph of minimum degree at least three contains a cycle whose length is a power of two. Carr recently proved that at least $4/7$ of the vertices in a minimal counterexample would have degree three. Write d for the number of cubic vertices and a for the number of vertices of degree at least four. We prove the uniform bound

$$d \geq 2a + 6.$$

Thus every minimal counterexample has at least $\lceil 2|V(G)|/3 \rceil + 2$ cubic vertices. The key object is a graph on the high-degree vertices obtained by suppressing the cubic vertices with two high-degree neighbours. Four-cycle exclusion and minimality make this incidence kernel simple, C_4 -free, and 2-degenerate; a sharp elementary edge bound and parity then give the result.

1 Introduction

The Erdős–Gyárfás conjecture asks whether every finite graph of minimum degree at least three contains a simple cycle of length 2^k for some $k \geq 2$. Throughout, graphs are finite, simple, and undirected, and all cycles are simple.

Following Carr [1], call a counterexample *minimal* if it is chosen with minimum possible order and, subject to that, minimum possible size. Carr proved that at least $4/7$ of its vertices are cubic. His proof records two useful consequences of minimality: the vertices of degree at least four are independent, and the cubic vertices dominate the graph. We reprove those observations and retain more of the resulting incidence structure.

Theorem 1. *Let G be a minimal counterexample to the Erdős–Gyárfás conjecture. Put*

$$D = \{v \in V(G) : d_G(v) = 3\}, \quad A = \{v \in V(G) : d_G(v) \geq 4\},$$

and write $d = |D|$, $a = |A|$. Then

$$d \geq 2a + 6.$$

Consequently, writing $n = |V(G)|$,

$$d \geq \left\lceil \frac{2n+6}{3} \right\rceil = \left\lceil \frac{2n}{3} \right\rceil + 2, \quad a \leq \left\lfloor \frac{n-6}{3} \right\rfloor.$$

*AI-assisted mathematical research campaign. The proof was independently derived and adversarially checked by separate OpenAI Codex agents; all mathematical claims remain the author’s responsibility.

2 The incidence kernel

We begin with the minimality facts, included to make the argument self-contained.

Lemma 2. *Every proper subgraph H of G has $\delta(H) \leq 2$. Moreover, A is independent and every vertex of G has a neighbour in D .*

Proof. If a proper subgraph H had minimum degree at least three, it would be a counterexample of smaller order, or of the same order and smaller size. If an edge joined two vertices of A , deleting it would preserve minimum degree at least three. Hence A is independent. Finally, if a vertex v had no neighbour in D , deleting v would lower each of its neighbours from degree at least four to degree at least three, again producing a smaller counterexample. $\square$

Every vertex of D therefore has at most two neighbours in A . For $i \in \{0, 1, 2\}$, let

$$D_i = \{v \in D : d_A(v) = i\}, \quad d_i = |D_i|.$$

Define a graph J on A by placing one edge between the two A -neighbours of each vertex of D_2 .

Lemma 3. *The incidence kernel J is simple, C_4 -free, and 2-degenerate.*

Proof. The two A -neighbours of a vertex in D_2 are distinct. Two different vertices of D_2 cannot have the same neighbour pair, because the two corresponding length-two paths would form a C_4 in G . Thus J is simple.

Every simple cycle of length ℓ in J lifts, by replacing its edges with their corresponding distinct two-edge paths, to a simple cycle of length 2ℓ in G . Hence J , and every subgraph of J , has no cycle of power-of-two length. If a subgraph of J had minimum degree at least three, it would be a counterexample on at most $a < n$ vertices. This contradicts the order-minimality of G . Therefore J is 2-degenerate. In particular it is C_4 -free. $\square$

We need the following small extremal fact.

Lemma 4. *If a simple C_4 -free, 2-degenerate graph has $r \geq 4$ vertices, then it has at most $2r - 4$ edges. If $r \geq 6$, it has at most $2r - 5$ edges.*

Proof. Every simple 2-degenerate graph on $r \geq 2$ vertices has at most $2r - 3$ edges. Equality forces every deletion in a degeneracy ordering to remove a vertex of degree exactly two, eventually leaving a triangle. Reversing the next deletion adds a vertex adjacent to two vertices of that triangle, which creates a C_4 . This proves the first bound.

Suppose now that $r \geq 6$ and equality $|E| = 2r - 4$ holds in the first bound. The same deletion argument reduces an equality example to the unique four-vertex, four-edge C_4 -free graph, the paw, and then to the unique five-vertex, six-edge example, two triangles sharing one vertex. Every pair of vertices in the latter graph has a common neighbour. Adding the next degree-two vertex therefore creates a C_4 , a contradiction. $\square$

3 Preliminary abundance bounds

Lemma 5. *Every minimal counterexample satisfies $d \geq 2a + 4$. If $a \geq 6$, then $d \geq 2a + 6$.*

Proof. First assume $a \geq 4$. Since the edges of J correspond bijectively to the vertices of D_2 , Lemma 4 gives

$$d_2 \leq 2a - 4.$$

Let $e = e_G(A, D)$ and introduce the nonnegative incidence slacks

$$x = e - 4a, \quad y = 2d - e = d_1 + 2d_0.$$

Since $e = d_1 + 2d_2$, we have $d_1 \geq 8 + x$ and hence $y \geq 8 + x$. The exact identity

$$2(d - 2a) = x + y$$

now yields $d \geq 2a + 4$.

If $a \geq 6$, the second part of Lemma 4 instead gives $d_2 \leq 2a - 5$, so $d_1 \geq 10 + x$ and $y \geq 10 + x$. Thus $d \geq 2a + 5$. Equality would force $x = 0$ and $y = 10$. But the degree sum inside $G[D]$ is

$$\sum_{v \in D} d_{G[D]}(v) = 3d - e = d + y,$$

which would be odd when $d = 2a + 5$ and $y = 10$. This contradicts the handshake lemma, and proves $d \geq 2a + 6$.

It remains to establish the universal bound for $a \leq 3$. If $a = 0$, then G is a nonempty simple cubic graph, so $d \geq 4$. If $a = 1$ and $d \leq 5$, the sole A -vertex has at least four distinct neighbours in D . For $d = 4$, the graph $G[D]$ is a four-cycle. For $d = 5$, parity forces all five vertices of D to meet A , so $G[D]$ is a five-cycle; the A -vertex together with any two-edge segment of that cycle gives a C_4 . Hence $d \geq 6$.

If $a = 2$, simplicity and C_4 -freeness give $d_2 \leq 1$. Thus

$$8 \leq d_1 + 2d_2 \leq d + 1.$$

The only value below $2a + 4 = 8$ is $d = 7$, which forces $(d_0, d_1, d_2) = (0, 6, 1)$. The degree sum in $G[D]$ is then $2d_1 + d_2 = 13$, impossible. Similarly, if $a = 3$, then $d_2 \leq 3$ and

$$12 \leq d_1 + 2d_2 \leq d + 3.$$

The only value below $2a + 4 = 10$ is $d = 9$, forcing $(d_0, d_1, d_2) = (0, 6, 3)$ and the odd internal degree sum $2d_1 + d_2 = 15$. This completes the proof. $\square$

4 Excluding the remaining equality profiles

It remains to rule out $d = 2a + 4$ and $d = 2a + 5$ when $a \leq 5$. Write $R = G[D]$ and colour each vertex of D_1 by its unique neighbour in A .

Lemma 6. *If two vertices of D_1 are the endpoints of a two-edge path in R , then their colours are distinct and have no common neighbour in J .*

Proof. If $u - z - w$ is such a path and u, w have the same colour α , then α, u, z, w form a four-cycle. Suppose instead that their distinct colours α, β have a common J -neighbour γ . Let $p, q \in D_2$ encode the kernel edges $\alpha\gamma$ and $\beta\gamma$. Then

$$\alpha, u, z, w, \beta, q, \gamma, p, \alpha$$

is an eight-cycle. Its vertices are distinct; in particular z cannot be p or q , because vertices of D_2 have degree one in R . $\square$

For $s = d - 2a \in \{4, 5\}$, the incidence identities take the form

$$x + y = 2s, \quad d_2 = 2a - s + d_0 + x, \quad d_1 = 2s - x - 2d_0. \quad (1)$$

The degree sum $d + y$ in R is even. Thus x is even when $s = 4$ and odd when $s = 5$.

Completion of the proof of Theorem 1. Lemma 5 already excludes both values of s for $a \geq 6$. We treat $a \leq 5$.

If $a = 0$, the case $s = 4$ is the cubic graph on four vertices, namely K_4 , and the case $s = 5$ contradicts the handshake lemma. Both are impossible.

Let $a = 1$. Since $d_2 = 0$, equation (1) leaves, for each s , either a 2-regular graph R all of whose vertices have the same colour, or $d_0 = 2$ and respectively four or five vertices in D_1 . The first alternative contains a two-edge segment closing through the sole A -vertex to a C_4 . In the second, the two-element R -neighbourhoods of the D_1 vertices must be pairwise disjoint by Lemma 6; respectively eight elements cannot fit in six vertices, and ten cannot fit in seven.

Let $a = 2$. Since J is simple, equation (1) leaves two profiles for each s . If J is empty, then R is 2-regular on eight or nine coloured vertices. On a cycle of length ℓ , adjacency at distance two gives one odd cycle for odd ℓ and two cycles of length $\ell/2$ for even ℓ . It is bipartite only when 4 divides ℓ . Avoiding a same-coloured distance-two pair would therefore make every component length divisible by four, forcing a forbidden C_4 or C_8 .

In the other profile, J is one edge and R has one degree-three vertex z , one degree-one vertex $e \in D_2$, and all remaining vertices in D_1 of degree two. Its non-cycle component is unicyclic, consisting of a cycle through z and a pendant path from z to e ; any other component would be a coloured cycle and is excluded by the preceding distance-two argument. Since e is adjacent to both vertices of A , the pendant path has exactly one coloured internal vertex: zero or at least two would immediately give a C_4 . For $s = 4$, the cycle is a six-cycle containing five coloured vertices; three alternating coloured vertices are pairwise at distance two, so two have the same colour and give a C_4 . For $s = 5$, write the cycle as

$$z, c_1, c_2, c_3, c_4, c_5, c_6, z.$$

The distance-two constraints on its coloured vertices form the path $c_2, c_4, c_6, c_1, c_3, c_5$. If the two colour classes on these six vertices are unbalanced, this path is not properly coloured and gives a C_4 . If they are balanced, proper colouring forces alternation, so c_2, c_3 have the same colour. Their complementary six-edge path around the seven-cycle, together with the two edges through their common A -neighbour, is a C_8 .

It remains to treat $a = 3, 4, 5$. Equations (1), simplicity, and the kernel edge bounds leave exactly the kernels P_3 , K_3 , the paw, and the five-vertex friendship graph F_2 . For $s = 4$ their values of d_1 are respectively 8, 6, 8, 8; for $s = 5$ they are 9, 7, 9, 9.

For $J = P_3$, call the colours L, M, R in path order. Let h be the number of D_1 vertices with two D_1 neighbours in R . The two degree-one vertices of D_2 give

$$h \geq 6 \quad (s = 4), \quad h \geq 7 \quad (s = 5).$$

Lemma 6 forces the two neighbour colours of each such vertex to be M and one of L, R . There are at most two M -coloured vertices for $s = 4$, and at most three for $s = 5$, with total R -degree at most four and six respectively. Both inequalities are impossible.

For $J = K_3$, Lemma 6 says no vertex of R has two D_1 neighbours. Each of the six or seven vertices of D_1 therefore needs a neighbour outside D_1 . The three degree-one vertices of D_2 supply three incidences, while the sole degree-three vertex of D_0 can supply at most

one without becoming the middle of a forbidden two-edge path. The total capacity four is insufficient.

For the paw, the four degree-one vertices of D_2 imply $h \geq 4$ for $s = 4$ and $h \geq 5$ for $s = 5$. The only distinct colour pair without a common J -neighbour is the degree-three centre and the leaf. Thus every counted vertex must be incident with a centre-coloured vertex. There are respectively at most one and two such vertices, of total R -degree at most two and four, again a contradiction.

Finally, in F_2 any two distinct colours occurring on D_1 have a common J -neighbour, including the shared centre if it receives the one extra colour in the $s = 5$ profile. Hence no vertex of R has two D_1 neighbours. All eight or nine degree-two vertices of D_1 require a neighbour in D_2 , but the six degree-one vertices of D_2 supply only six incidences.

We have excluded $s = 4, 5$ in every remaining case. Together with Lemma 5, this proves $d \geq 2a + 6$. Substituting $a = n - d$ gives $3d \geq 2n + 6$ and $3a \leq n - 6$, which are exactly the stated integer bounds. $\square$

5 A related size bound

The same minimality framework gives a useful accompanying estimate.

Corollary 7. *If $m = |E(G)|$, then $m \leq 2n - 3$.*

Proof. Choose a cubic vertex v . Every subgraph of $G - v$ is a proper subgraph of G , so $G - v$ is 2-degenerate. Hence

$$m - 3 = |E(G - v)| \leq 2(n - 1) - 3,$$

which gives $m \leq 2n - 2$.

If equality held, G would be an n -vertex, $(2n - 2)$ -edge graph with no proper subgraph of minimum degree three. Narins, Pokrovskiy and Szabó [2, Theorem 1.4] proved that every such graph is pancyclic. In particular it contains a C_4 , a contradiction. $\square$

6 Verification and priority boundary

The proof was independently derived and then checked line by line by a separate agent. The kernel edge bounds were also exhaustively tested on every labelled graph through seven vertices; this computation is an audit aid, not a premise. Searches on 3 August 2026 for the exact constants and the incidence-kernel construction found Carr’s 4/7 theorem but no earlier statement of the bounds above. That is a search-relative novelty check, not a guarantee concerning unpublished or unindexed work. The theorem is a necessary condition on a hypothetical counterexample and does not resolve the Erdős–Gyárfás conjecture.

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References

- [1] A. Carr, *Every Minimal Counterexample to the Erdős–Gyárfás Conjecture is Predominantly Cubic*, arXiv:2605.22844, 2026. <https://arxiv.org/abs/2605.22844>.
- [2] L. Narins, A. Pokrovskiy and T. Szabó, *Graphs without proper subgraphs of minimum degree 3 and short cycles*, *Combinatorica* **37** (2017), 495–519. <https://arxiv.org/abs/1408.5289>.